

# CSI3015: Software Project Management

## The COCOMO Model

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# What is the COCOMO Model?

## Definition

**COCOMO** (Constructive Cost Model) is a procedural software cost estimation model developed by **Barry W. Boehm** in 1981.

- ▶ Used to estimate **effort**, **cost**, and **schedule** for a software project.
- ▶ Based on the size of the software, typically measured in estimated lines of code (LOC or KLOC).
- ▶ Built on regression formulas derived from historical project data.
- ▶ Categorizes projects based on complexity and team environment.

# The Three Modes of COCOMO

Boehm categorized software projects into three distinct modes based on their complexity and the development environment:

## Organic

- ▶ Small, highly experienced teams.
- ▶ Familiar, relaxed environment.
- ▶ Well-understood requirements.
- ▶ *Ex: Simple payroll system.*

## Semi-detached

- ▶ Medium teams.
- ▶ Mix of experienced and new members.
- ▶ Moderate complexity.
- ▶ *Ex: OS utility, DB design.*

## Embedded

- ▶ Tight hardware/software constraints.
- ▶ Unique, highly complex problems.
- ▶ *Ex: ATM, Flight control software.*

# The Three Levels of COCOMO

To provide varying levels of accuracy depending on the project phase, COCOMO is divided into three levels:

1. **Basic COCOMO:**

Used for quick, early, and rough estimations based solely on estimated lines of code. Best for small-to-medium-sized projects.

2. **Intermediate COCOMO:**

Expands on the basic model by introducing 15 "Cost Drivers" (e.g., personnel skill, hardware constraints) to adjust the base estimation for accuracy.

3. **Detailed COCOMO:**

The most comprehensive level. It assesses the impact of all cost drivers at each specific phase of the engineering process (planning, design, coding, testing).

# Basic COCOMO: Key Formulas

The Basic COCOMO model calculates effort and development time using established regression equations:

## 1. Effort ( $E$ )

$$E = a_b \times (\text{KLOC})^{b_b}$$

*(Measured in Person-Months)*

## 2. Development Time ( $D$ )

$$D = c_b \times (E)^{d_b}$$

*(Measured in Months)*

## 3. Average Staffing Size ( $SS$ )

$$SS = \frac{E}{D}$$

*(Measured in Persons)*

# Basic COCOMO: Constants Matrix

The constants  $a_b$ ,  $b_b$ ,  $c_b$ , and  $d_b$  vary depending on the project mode:

| Project Mode  | $a_b$ | $b_b$ | $c_b$ | $d_b$ |
|---------------|-------|-------|-------|-------|
| Organic       | 2.4   | 1.05  | 2.5   | 0.38  |
| Semi-detached | 3.0   | 1.12  | 2.5   | 0.35  |
| Embedded      | 3.6   | 1.20  | 2.5   | 0.32  |

# Going Beyond Basic Estimations

## Intermediate COCOMO

Calculates an **Effort Adjustment Factor (EAF)** by rating 15 Cost Drivers across four categories:

- ▶ **Product:** Reliability, Database size, Complexity.
- ▶ **Hardware:** Execution time, Memory constraints.
- ▶ **Personnel:** Analyst/Programmer capability, Experience.
- ▶ **Project:** Software tools, Schedule constraints.

*New Effort Formula:  $E = a_i \times (KLOC)^{b_i} \times EAF$*

## Detailed COCOMO

Applies these cost driver multipliers phase-by-phase (e.g., multipliers for the design phase differ from testing phase multipliers), yielding the highest accuracy.

# Intermediate COCOMO: The 15 Cost Drivers (Part 1)

To calculate the Effort Adjustment Factor (EAF), each of these 15 drivers is rated from Very Low to Extra High:

## 1. Product Attributes

- ▶ **RELY:** Required software reliability
- ▶ **DATA:** Size of application database
- ▶ **CPLX:** Complexity of the product

## 2. Hardware (Computer) Attributes

- ▶ **TIME:** Run-time performance constraint
- ▶ **STOR:** Main memory constraint
- ▶ **VIRT:** Volatility of the virtual machine environment
- ▶ **TURN:** Required turnaround time

# Intermediate COCOMO: The 15 Cost Drivers (Part 2)

## 3. Personnel Attributes

- ▶ **ACAP:** Analyst capability
- ▶ **AEXP:** Application experience
- ▶ **PCAP:** Programmer capability
- ▶ **VEXP:** Virtual machine experience
- ▶ **LEXP:** Programming language experience

## 4. Project Attributes

- ▶ **MODP:** Use of modern programming practices
- ▶ **TOOL:** Use of software tools
- ▶ **SCED:** Required development schedule

# Summary

- ▶ **COCOMO** provides a quantifiable, algorithmic foundation for project planning and resource allocation.
- ▶ Project managers must correctly identify the **Mode** (Organic, Semi-detached, Embedded) based on team constraints and project complexity.
- ▶ The **Level** of estimation (Basic, Intermediate, Detailed) should evolve as the project transitions from early conceptualization into detailed execution.
- ▶ While Agile methodologies often rely on story points today, parametric models like COCOMO remain crucial for large-scale, enterprise, and legacy engineering bids.